

APPROXIMATIONS FOR COLLISIONAL AND RADIATIVE TRANSITION RATES IN ATOMIC HYDROGEN

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ABSTRACT

A simple, accurate, and complete set of approximate formulae is presented for evaluating collisional and radiative transition rates for bound-bound and bound-free transitions in atomic hydrogen. The approximations are chosen in a form which facilitates computer programming and solution of nonequilibrium excitation problems of interest in plasma physics and astrophysics. Detailed comparisons are made with experimental, theoretical, and other semiempirical results.

I. INTRODUCTION

The distribution of atomic hydrogen among its various levels when local thermodynamic equilibrium cannot be assumed is a recurrent problem in plasma physics and astrophysics. Although the problem has been treated in considerable detail, especially by Bates, Kingston, and McWhirter (1962*a, b*; Bates and Kingston 1963), cases continue to arise for which their tables are either not sufficiently extensive or for which they are inaccurate. For example, it has been shown (Johnson and Hinnov 1969) that their recombination rate coefficients for low-temperature plasmas are too large. This is a consequence of using classical cross-sections (Gryzinski 1959) which, for strong transitions among excited levels, are too large at low electron energies (Seaton 1962; Saraph 1964). Furthermore, their tables do not extend to the high temperatures encountered recently in Tokamak plasmas. Moreover, since the classical cross-sections are too small at high impact energies (Seaton 1962; Saraph 1964), a simple extension of their calculations to higher temperatures may give erroneous results.

Numerous semiempirical formulae have been proposed for representing excitation and ionization cross-sections (e.g., Drawin 1967; Sampson 1969; Park 1971). Although many of these expressions reproduce measured cross-sections to within experimental error, they usually apply only to transitions involving the ground level. In most cases, the formulae cannot be integrated in closed form over a Maxwellian electron energy distribution to obtain rate coefficients, so that one must resort to numerical integration or to additional approximations, which may be valid only for a narrow range of electron temperatures.

In view of these considerations, we present semiempirical formulae for electron-impact cross-sections for atomic hydrogen which are more accurate than the classical cross-sections usually used and by means of which the nonequilibrium excitation problem may be easily programmed for computer solution. To simplify the programming of such problems, we also present approximate formulae for bound-bound and bound-free radiative transitions which permit the evaluation of radiative transition rates without the use of extensive tables (Green, Rush, and Chandler 1957; Seaton 1959).

II. RADIATIVE TRANSITIONS

The absorption oscillator strength for a transition from a lower level with principal quantum number n to an upper level n' is given approximately by (Bethe and Salpeter 1957)

$$f_{nn'} \approx \frac{32}{3(\sqrt{3})\pi} \frac{n}{n'^3} x^{-3}, \quad (1)$$

where x is the ratio of the transition energy to the ionization energy of the lower level

$$x = E_{nn'}/I_n = 1 - (n/n')^2. \quad (2)$$

If we introduce a correction factor $g(n, x)$ such that equation (1) becomes exact,

$$f_{nn'} = \frac{32}{3(\sqrt{3})\pi} \frac{n}{n'^3} x^{-3} g(n, x), \quad (3)$$

then $g(n, x)$ is continuous across the ionization limit, becoming the usual Gaunt factor for bound-free transitions.

We may approximate $g(n, x)$ with sufficient accuracy for most purposes by a simple polynomial in x :

$$g(n, x) = g_0(n) + g_1(n)x^{-1} + g_2(n)x^{-2}. \quad (4)$$

The oscillator strengths of Green *et al.* (1957) for $1 \leq n \leq 19$ and $n+1 \leq n' \leq 20$ may be reproduced to within 0.5 percent by using equations (3) and (4) with the coefficients in table 1. This table was obtained by finding $g_0(n)$, $g_1(n)$, and $g_2(n)$ such that quantum-mechanical results (Green *et al.* 1957; Karzas and Latter 1961) are reproduced for $n' = n+1$, $n' = n+2$, and $n' \rightarrow \infty$, and then generating approximating formulae for these coefficients.

The rate coefficient for spontaneous radiative recombination to a level n may be written (Seaton 1959)

$$S_\nu(c, n) = \mathfrak{D}(I_n/kT)^{3/2} \exp(I_n/kT) \int_1^\infty \exp[-(I_n/kT)x] g(n, x) x^{-1} dx, \quad (5)$$

where $g(n, x)$ is the Gaunt factor, x is the ratio of transition energy to ionization energy, and

$$\mathfrak{D} = (64/3)(\pi/3)^{1/2} \alpha^4 a_0^2 c = 5.197 \times 10^{-14} \text{ cm}^3 \text{ s}^{-1}. \quad (6)$$

Here α is the fine-structure constant, a_0 is the Bohr radius, and c is the speed of light. Tables of $S_\nu(c, n)$ have been prepared by Seaton (1959), who used the first three terms of an asymptotic expansion (Menzel and Pekeris 1935; Burgess 1958) for $g(n, x)$. Alternatively, $g(n, x)$ may be approximated by using equation (4) and table 1. Equation (5) may then be easily integrated to yield

$$S_\nu(c, n) = \mathfrak{D}(I_n/kT)^{3/2} \exp(I_n/kT) \sum_{i=0}^2 g_i(n) E_{i+1}(I_n/kT), \quad (7)$$

where E_i is the exponential integral

$$E_i(z) = \int_1^\infty e^{-zt} t^{-i} dt \quad (i = 0, 1, 2, \dots). \quad (8)$$

TABLE 1
GAUNT FACTOR COEFFICIENTS

	$n=1$	$n=2$	$n \geq 3$
$g_0(n)$	1.1330	1.0785	$0.9935 + 0.2328 n^{-1} - 0.1296 n^{-2}$
$g_1(n)$	-0.4059	-0.2319	$-n^{-1}(0.6282 - 0.5598 n^{-1} + 0.5299 n^{-2})$
$g_2(n)$	0.07014	0.02947	$n^{-2}(0.3887 - 1.181 n^{-1} + 1.470 n^{-2})$

The function $E_i(z)$ for $i \geq 1$ is readily obtained by using a polynomial approximation (Abramowitz and Stegun 1964) for $E_1(z)$ together with the recursion relation

$$E_{i+1}(z) = \frac{1}{i} [e^{-z} - zE_i(z)] \quad (i = 1, 2, 3, \dots) \quad (9)$$

Recombination rate coefficients obtained from equation (7) are comparable in accuracy with those derived from Seaton's (1959) tables. Figure 1 shows ratios of rate coefficients obtained from equation (7) to those obtained by piecewise integration using the tabulated bound-free Gaunt factors of Karzas and Latter (1961). For temperatures less than 10^6 K, equation (7) gives results accurate to within 5 percent.

III. BETHE APPROXIMATION COEFFICIENTS

Bethe (1930) has shown that for large impact energies E , the cross-section for an electron-induced, bound-bound transition may be written

$$Q_{nn'} \sim \frac{2I_H}{E} \left(A_{nn'} \ln \frac{E}{2I_H} + B_{nn'} \right) \pi a_0^2, \quad (10)$$

where $I_H = 13.6$ eV and $A_{nn'}$ and $B_{nn'}$ are dimensionless coefficients. The coefficient $A_{nn'}$ may be written in our present notation as

$$A_{nn'} = 2n^2 x^{-1} f_{nn'}. \quad (11)$$

The coefficient $B_{nn'}$ is usually obtained by requiring that equation (10) agree with the first Born approximation at high impact energies (Moiseiwitsch and Smith 1968).

For ionization from level n the cross-section per unit range of energy ϵ of the ejected electron is

$$\frac{dQ_n}{d\epsilon} \sim \frac{2I_H}{E} \left(\frac{dA_n}{d\epsilon} \ln \frac{E}{2I_H} + \frac{dB_n}{d\epsilon} \right) \pi a_0^2. \quad (12)$$

The variables ϵ and x are related by

$$x = 1 + \epsilon/I_n, \quad (13)$$

so that equation (12) may be written

$$\frac{dQ_n}{dx} \sim \frac{2I_H}{E} \left(\frac{dA_n}{dx} \ln \frac{E}{2I_H} + \frac{dB_n}{dx} \right) \pi a_0^2. \quad (14)$$

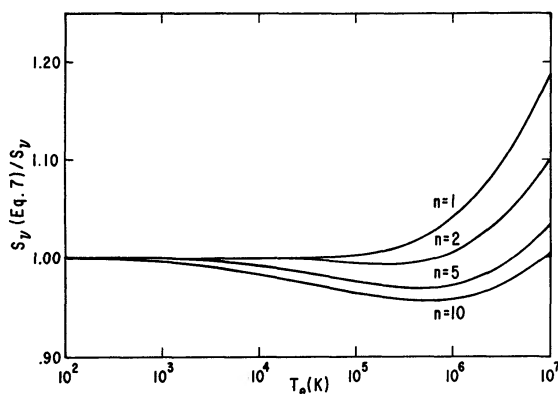


FIG. 1.—A comparison of rate coefficients for direct radiative recombination to level n , obtained from eq. (7), with those obtained by piecewise integration using Gaunt factors of Karzas and Latter.

Then the total ionization cross-section becomes, upon integration of equation (14),

$$Q_n \sim \frac{2I_H}{E} \left(A_n \ln \frac{E}{2I_H} + B_n \right) \pi a_0^2, \quad (15)$$

where

$$A_n = \int_1^\infty \frac{dA_n}{dx} dx \quad (16)$$

and

$$B_n = \int_1^\infty \frac{dB_n}{dx} dx. \quad (17)$$

A simple approximation for A_n may be obtained by combining equations (3), (4), and (11), and noting that, for excitation to bound levels with large n' ,

$$dx = \frac{2n^2}{n'^3} dn'. \quad (18)$$

Then the integrand of equation (16) becomes

$$\frac{dA_n}{dx} = \frac{32}{3(\sqrt{3})\pi} n x^{-4} \sum_{i=0}^2 g_i(n) x^{-i}, \quad (19)$$

and integration yields

$$A_n = \frac{32}{3(\sqrt{3})\pi} n \sum_{i=0}^2 \frac{g_i(n)}{i+3}. \quad (20)$$

Equation (20) and table 1 give values of A_n which agree to within 0.5 percent with those of Kingston (1965) for $1 \leq n \leq 6$.

Omidvar (1969a) has given an expansion for dB_1/dx in terms of ϵ which is valid for large ϵ . In the present notation, this expansion becomes, when referred to the n th initial level,

$$\frac{dB_n}{dx} \sim 2n^2 x^{-2} (1 + \frac{4}{3} x^{-1}). \quad (21)$$

This expression, when multiplied by $(2I_H/E) \pi a_0^2$, is precisely the differential cross-section given by classical theory (Burgess and Percival 1968) when the bound electron is assumed to have a kinetic energy equal to its binding energy. By adding another term with an adjustable coefficient, we obtain a satisfactory polynomial approximation for $B_{nn'}$ and dB_n/dx , hence B_n :

$$\frac{dB_n}{dx} = 2n^2 x^{-2} (1 + \frac{4}{3} x^{-1} + b_n x^{-2}), \quad (22)$$

$$B_{nn'} = \frac{4n^4}{n'^3} x^{-2} (1 + \frac{4}{3} x^{-1} + b_n x^{-2}), \quad (23)$$

$$B_n = \frac{2}{3} n^2 (5 + b_n). \quad (24)$$

We choose b_n to minimize the average error of $B_{nn'}$ for $n' = n + 1$ and $n' = n + 2$:

$$b_1 = -0.603, \quad (25)$$

$$b_n = n^{-1} (4.0 - 18.63n^{-1} + 36.24n^{-2} - 28.09n^{-3}), \quad n \geq 2. \quad (26)$$

Equation (23) then gives results for $B_{nn'}$ which agree to within 3 percent with Born-approximation values for $1 \leq n \leq 9$ (Kingston and Lauer 1966a, b ; Omidvar 1969b), and B_1 agrees to within 1 percent with the value given by Omidvar (1969a).

Bethe coefficients and oscillator strengths may be reproduced with greater precision by including additional terms in equations (4) and (22)–(24), of course, but at the expense of a considerable increase in complexity of the resulting cross-section formulae.

It should perhaps be pointed out that the numerical values of the Bethe coefficients depend upon the form chosen for expressing the asymptotic cross-section. Kingston (1965) and Kingston and Lauer (1966*a, b*) use common logarithms while Omidvar (1969*a, b*) uses I_H rather than $2I_H$ as the unit of energy. Thus the values of A_1 and B_1 given by Omidvar (1969*a*) become 0.566825 and 2.90602, respectively, when expressed in the present units.

IV. EXCITATION CROSS-SECTIONS

Although the first Born approximation, and therefore the Bethe approximation, gives reliable results at high impact energies, it fails at low and intermediate energies. In constructing a semiempirical formula valid at all energies, it is necessary to consider other theoretical results as well as experimental measurements.

For excitation and ionization from the ground level of atomic hydrogen there is generally satisfactory agreement between theory and experiment. Information concerning transitions not involving the ground level is much more sketchy. While no direct measurements of these cross-sections exist, the impact-parameter approximation (Seaton 1962; Saraph 1964), which should apply for strong dipole transitions, sets rather definite upper limits on the magnitudes for $n \rightarrow n+1$ transitions when $n > 2$. These predictions have been confirmed by Johnson and Hinnov (1969) for $3 \lesssim n \lesssim 8$ by comparing helium recombination spectra in the C stellarator with detailed computations in which rate coefficients were adjusted to obtain best agreement.

For $n' - n > 1$ the impact-parameter approximation does not apply quantitatively, but it suggests that the cross-sections for such transitions at low impact energies will more closely approach the Born-approximation values than will those for $\Delta n = 1$. Since the $n \rightarrow n+1$ transitions are the dominant ones, moderate errors in $\Delta n > 1$ cross-sections may be tolerated without seriously affecting the accuracy of level-population calculations.

We seek a formula which accurately represents the most reliable cross-sections for $n \rightarrow n'$ transitions in atomic hydrogen, which results in easily evaluated rate coefficients, and which leads to an equally acceptable formula for ionization cross-sections.

Let $U_{nn'}$ be the normalized impact energy,

$$U_{nn'} = E/E_{nn'}. \quad (27)$$

Then the Bethe approximation, equation (10), may be written

$$Q_{nn'} \sim \frac{2n^2}{x} U_{nn'}^{-1} \left[A_{nn'} \ln U_{nn'} + \left(B_{nn'} - A_{nn'} \ln \frac{2n^2}{x} \right) \right] \pi a_0^2. \quad (28)$$

We regard equation (28) as the first two terms of an expansion in $U_{nn'}$. By adding a $U_{nn'}^{-2}$ term to equation (28) and introducing an additional factor which approaches unity for large $U_{nn'}$ and which contains an adjustable parameter, we arrive at a satisfactory representation.

$$Q_{nn'} = \frac{2n^2}{x} U_{nn'}^{-1} [1 - \exp(-r_{nn'} U_{nn'})] \left[A_{nn'} \left(\ln U_{nn'} + \frac{1}{2U_{nn'}} \right) + \left(B_{nn'} - A_{nn'} \ln \frac{2n^2}{x} \right) \left(1 - \frac{1}{U_{nn'}} \right) \right] \pi a_0^2. \quad (29)$$

The factor containing the parameter $r_{nn'}$ reduces the cross-section at low energies if $r_{nn'}$ is small. As mentioned above, we expect this reduction to be greater for $\Delta n = 1$. For this reason, and to simplify the deduction of a formula for ionization cross-sections,

we shall assume

$$r_{nn'} = r_n x. \quad (30)$$

We then require for each level a single parameter r_n which enters into all excitation cross-sections and the ionization cross-section involving that level. As we shall see, our requirements are met by

$$r_1 = 0.45, \quad (31)$$

$$r_n = 1.94n^{-1.57}, \quad n \geq 2. \quad (32)$$

Cross-sections obtained from equations (29)–(32) are compared in figures 2–7 with various experimental, theoretical, and semiempirical results. The figures show the Born-approximation calculations of Omidvar (1965b), the impact-parameter approximation calculations of Saraph (1964), six-state close-coupling calculations of Burke, Ormonde, and Whitaker (1967), classical cross-sections obtained from a formula given by Bates *et al.* (1962a), and semiempirical results of Sampson and Golden (1970) and of Johnson and Hinnov (1969). In addition, figure 2 shows pseudo-state calculations of Burke and Webb (1970) and a composite of experimental results (Long, Cox, and Smith 1968; McGowan, Williams, and Curley 1969; Kauppila, Ott, and Fite 1970) for the $1 \rightarrow 2$ transition, and figure 3 shows measurements of Kleinpoppen and Kraiss (1967) for the $1 \rightarrow 3$ transition, normalized to the Born approximation.

For the $1 \rightarrow 2$ transition shown in figure 2 the close-coupling, pseudo-state, and impact-parameter results are in good agreement with one another and with experiment and are well represented by equations (29)–(31). The sharp peak in the experimental results for the $1 \rightarrow 3$ transition shown in figure 3 is not reproduced by any of the approximate cross-sections shown but should not have a serious effect on rate coefficients. No direct or indirect experimental evidence is available for the $2 \rightarrow 3$ transition shown in figure 4, but the close-coupling and impact-parameter results are in fairly close agreement and should be reliable. Figures 5–7 illustrate the close agreement between impact-parameter results for $n > 2$ and the optimized cross-sections deduced from recombination spectra. Although these results are not precise, they are not likely to be in error by a factor of 2 and are certainly more accurate at low and intermediate energies than the classical cross-sections or the Born approximation.

The rather complicated formula proposed by Sampson and Golden (1970) accurately reproduces the Born approximation at high energies and gives fairly reliable results at low energies for $n \leq 2$. For $n > 2$, however, their results at low energies are considerably too large. Furthermore, their formula leads to rate coefficients involving noninteger-order exponential integrals, which are not suitable for rapid computer evaluation.

Except for the work of Mihalas (1967), which used a formula given by Johnson (1967), the approximations reviewed by Drawin (1967), Sampson (1969), and Park (1971) likewise overestimate cross-sections at low energies for $n > 2$ and $\Delta n = 1$.

V. IONIZATION CROSS-SECTIONS

An approximation for ionization cross-sections may be generated from equation (29), our excitation cross-section approximation. Let U_n be the impact energy normalized with respect to the ionization energy.

$$U_n = E/I_n. \quad (33)$$

Then, on replacing $U_{nn'}$ by U_n/x and using equation (30), equation (29) may be converted into a differential ionization cross-section:

$$\begin{aligned} \frac{dQ_n}{dx} = \frac{2n^2}{U_n} [1 - \exp(-r_n U_n)] & \left[\frac{dA_n}{dx} \left(\ln \frac{U_n}{x} + \frac{x}{2U_n} \right) \right. \\ & \left. + \left(\frac{dB_n}{dx} - \frac{dA_n}{dx} \ln \frac{2n^2}{x} \right) \left(1 - \frac{x}{U_n} \right) \right] \pi a_0^2. \quad (34) \end{aligned}$$

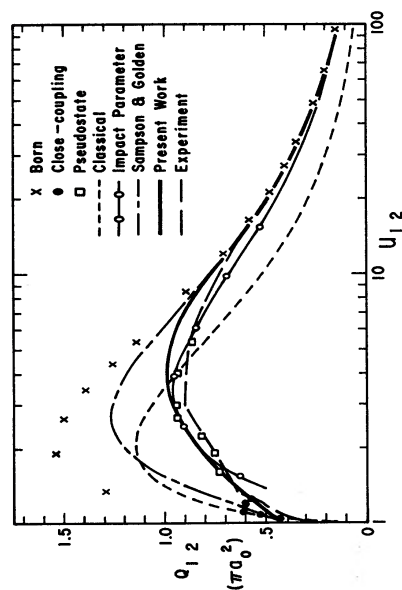


FIG. 2

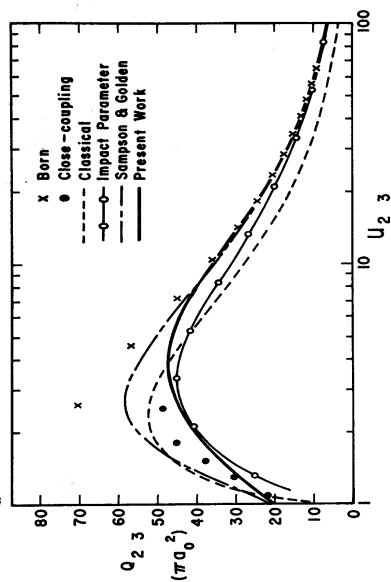


FIG. 4

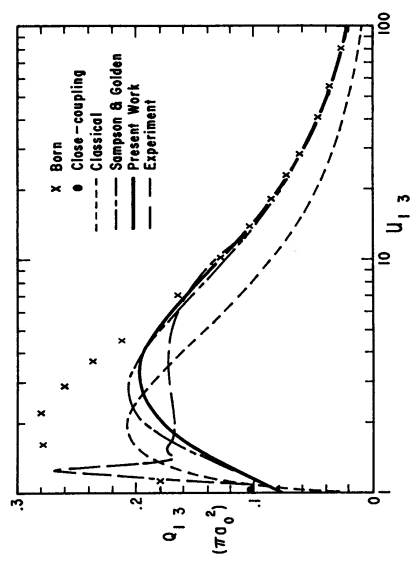


FIG. 3

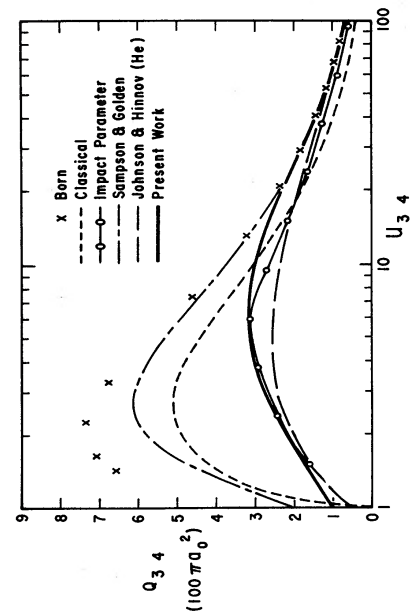


FIG. 5

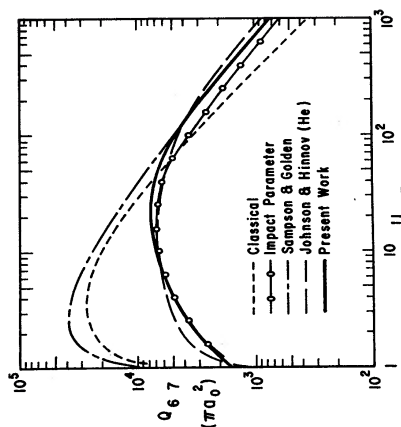


FIG. 6

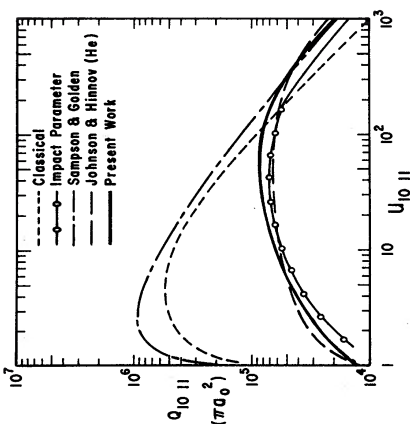


FIG. 7

FIG. 2.—Cross-sections for the $1 \rightarrow 2$ transition in atomic hydrogen. The experimental curve is a composite of results referred to in the text.

FIG. 3.—Cross-sections for the $1 \rightarrow 3$ transition in atomic hydrogen. The experimental curve is obtained from measurements of Kleinpoppen and Kraiss normalized to the Born approximation.

FIG. 4.—Cross-sections for the $2 \rightarrow 3$ transition in atomic hydrogen.

FIG. 5.—Cross-sections for the $3 \rightarrow 4$ transition in atomic hydrogen.

FIG. 6.—Cross-sections for the $6 \rightarrow 7$ transition in atomic hydrogen.

FIG. 7.—Cross-sections for the $10 \rightarrow 11$ transition in atomic hydrogen.

If equations (19) and (22) are substituted into equation (34), integration may be performed exactly to obtain the corresponding total ionization cross-section. The result is lengthy and will not be given here. However, the dominant terms may be expressed simply, in terms of the Bethe coefficients:

$$Q_n = \frac{2n^2}{U_n} [1 - \exp(-r_n U_n)] [A_n \ln U_n + (B_n - A_n \ln 2n^2)(1 - 1/U_n)^2] \pi a_0^2 \quad (35)$$

where A_n and B_n are given by equations (20) and (24), respectively.

In figure 8 the experimental cross-sections of Fite and Brackman (1958) for ionization from the ground level are compared with results of the Born approximation (Omidvar 1965a), the classical approximation (Gryzinski 1959), and the semiempirical formula of Sampson and Golden (1970), and with two cross-sections from the present work, one obtained from equation (35) and one from exact integration of equation (34). Except for the Born cross-sections being too large at low energies and the classical cross-sections too small at high energies, there is reasonable agreement among all the results shown.

The difference between equation (35) and exact integration of equation (34) illustrated in figure 8 is typical of that obtained for ionization from other levels. In view of the uncertainty of equation (34) at low energies and the simplicity of equation (35), the latter may be adopted as a satisfactory representation of ionization cross-sections.

VI. RATE COEFFICIENTS

Integration of equation (29) with a Maxwellian electron distribution is straightforward and gives the excitation rate coefficient

$$S_e(n, n') = (8kT/\pi m)^{1/2} \frac{2n^2}{x} \pi a_0^2 y^2 \left\{ A_{nn'} \left[\left(\frac{1}{y} + \frac{1}{2} \right) E_1(y) - \left(\frac{1}{z} + \frac{1}{2} \right) E_1(z) \right] \right. \\ \left. + \left(B_{nn'} - A_{nn'} \ln \frac{2n^2}{x} \right) \left[\frac{1}{y} E_2(y) - \frac{1}{z} E_2(z) \right] \right\}, \quad (36)$$

where

$$y = E_{nn'}/kT \quad (37)$$

and

$$z = r_{nn'} + E_{nn'}/kT. \quad (38)$$

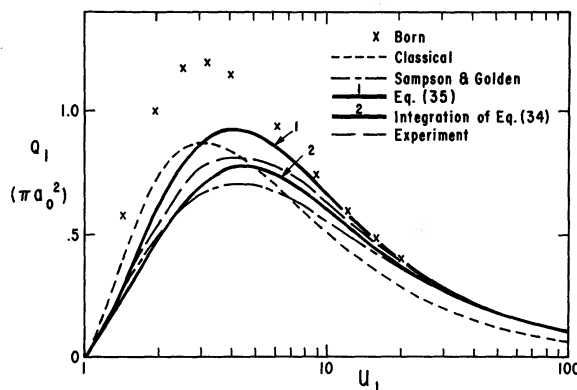


FIG. 8.—Cross-sections for ionization from the ground level of atomic hydrogen. The experimental curve represents measurements of Fite and Brackmann.

Similarly, the ionization rate coefficient obtained by integration of equation (35) is

$$S_e(n, c) = (8kT/\pi m)^{1/2} 2n^2 \pi a_0^2 y_n^2 \left\{ A_n \left[\frac{1}{y_n} E_1(y_n) - \frac{1}{z_n} E_1(z_n) \right] + (B_n - A_n \ln 2n^2) [\xi(y_n) - \xi(z_n)] \right\}, \quad (39)$$

where

$$y_n = I_n/kT, \quad (40)$$

$$z_n = r_n + I_n/kT, \quad (41)$$

and

$$\xi(t) = E_0(t) - 2E_1(t) + E_2(t). \quad (42)$$

VII. REDUCTION OF THE IONIZATION LIMIT

For nonequilibrium excitation problems in a plasma it is desirable, from both a physical and a practical standpoint, to regard high-lying bound states as an extension of the continuum rather than as discrete bound levels. Let n_0 be the principal quantum number of a level above which Saha equilibrium populations may be assumed. We consider an effective continuum c' composed of free states and all bound levels with $n > n_0$. The equations given above which relate to bound-free transitions may then be rewritten to refer to c' . The effective radiative-recombination rate coefficient becomes

$$S_r(c', n) = \mathfrak{D} \left(\frac{I_n}{kT} \right)^{3/2} \exp(I_n/kT) \sum_{i=0}^2 g_i(n) x_0^{-i} E_{i+1} \left(x_0 \frac{I_n}{kT} \right), \quad (7')$$

where

$$x_0 = 1 - (n/n_0)^2. \quad (43)$$

The Bethe coefficients become

$$A'_n = \frac{32}{3\sqrt{3}\pi} n \sum_{i=0}^2 \frac{g_i(n)}{i+3} x_0^{-(i+3)} \quad (20')$$

and

$$B'_n = \frac{2}{3} n^2 x_0^{-1} (3 + 2x_0^{-1} + b_n x_0^{-2}). \quad (24')$$

The effective ionization cross-section may be written

$$Q'_n = \frac{2n^2}{U'_n x_0} [1 - \exp(-r_n x_0 U'_n)] \left[A'_n \ln U'_n + \left(B'_n - A'_n \ln \frac{2n^2}{x_0} \right) \left(1 - \frac{1}{U'_n} \right)^2 \right] \pi a_0^2 \quad (35')$$

where

$$U'_n = U_n/x_0 = \frac{E}{I_n x_0}. \quad (33')$$

Finally, the effective ionization rate coefficient becomes

$$S_e(n, c') = (8kT/\pi m)^{1/2} \frac{2n^2}{x_0} \pi a_0^2 y_n'^2 \left\{ A'_n \left[\frac{1}{y_n'} E_1(y_n') - \frac{1}{z_n'} E_1(z_n') \right] + \left(B'_n - A'_n \ln \frac{2n^2}{x_0} \right) [\xi(y_n') - \xi(z_n')] \right\}, \quad (39')$$

where

$$y'_n = x_0 y_n = x_0 I_n / kT \quad (40')$$

and

$$z'_n = x_0 z_n = x_0 (r_n + I_n / kT). \quad (41')$$

Rate coefficients for collisional recombination and de-excitation may, of course, be obtained from detailed balance.

A revision and extension of the tables of Bates *et al.* (1962*a*, *b*, 1963), using the present results, is in preparation and will be published elsewhere.

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